

Eli Whitaker

Oakland, CA | (916) 284-3246 | eli.p.p.whitaker@gmail.com | eliwhitaker.org | github.com/rekatiwhile

SUMMARY

Mechanical Engineering graduate from UCLA (4.000 GPA, Summa Cum Laude) who integrates hardware and software into working physical systems, then builds the tooling to keep them debuggable. Brought two robots from bare parts to closed-loop autonomous operation, and when one had too many interacting variables to tune by hand-editing code, built a full-stack debug dashboard to make it diagnosable instead. Comfortable owning a problem across mechanical, electrical, and software boundaries.

EDUCATION

University of California, Los Angeles (UCLA)

Los Angeles, CA

B.S. Mechanical Engineering | GPA: 4.000 | Summa Cum Laude

June 2026

- Relevant Coursework: Robotics, Control of Robotic Systems, Dynamic Systems and Control, Mechatronics/Capstone Design, Electrical and Electronic Circuits, Circuit Measurements Lab.

TECHNICAL SKILLS

- **Systems Integration:** hardware/software interface validation, sensor-to-compute wiring, subsystem bring-up, calibration procedures, developer tooling for live tuning, root-cause debugging across mechanical/electrical/software layers.
- **Embedded / Interfaces:** Teensy 4.1 (AccelStepper, TMC2209 UART, INA219 current sensing), Jetson Orin, stepper motors/drivers, servo PWM, Dynamixel actuators, GRBL gantry control, UART/serial, CAN bus.
- **Software / Tools:** Python, C++, MATLAB, ROS 2, Flask, JavaScript, Git/GitHub, OpenCV, Linux CLI, serial communication scripting.
- **Hardware / Test:** prototype assembly, bench testing, trial logging, calibration workflows, 3D printing (Bambu Lab P1S), SolidWorks, Onshape.

RESEARCH EXPERIENCE

Structures-Computer Interaction Lab – Robotic Laser Weeding

Aug. 2025 – Jun. 2026

Undergraduate Researcher

UCLA

- Integrated a field robot platform: AgileX SCOUT 2.0 rover, CNC gantry, 60 W blue diode laser, Jetson compute, and stereo USB cameras into a working autonomous system.
- Calibrated the stereo-to-robot coordinate transform by driving an AprilTag on the end effector through a 3D grid of poses and fitting an affine transform plus height-indexed homography matrices from the logged data.
- Rewrote the stereo constellation matcher and added target re-identification to keep the system locked onto a plant across camera motion; supported system characterization across 70 autonomous trials and 525 planned treatments.

ENGINEERING PROJECTS

Robot Systems Integration & Debug Dashboard

SCI Lab – Laser Weeding Platform

- Built a Flask/JavaScript remote debugging tool after running into too many interacting variables to tune by hand-editing constants in code; the goal was to make the robot diagnosable and tunable during integration, not just a status page.
- Built live camera tuning controls (exposure, gain, white balance), YOLO debug tooling (crop size, confidence thresholds, class filters, burst count), and stereo matching/triangulation debug tools with 3D target visualization and CSV/JSON/PNG export.
- Built browser-based gantry controls (homing, jog/absolute moves, stop/reset/unlock, raw GRBL commands) and workspace reprojection tools to check calibration by projecting robot-frame coordinates back into camera images.
- Built cached survey/replay workflows and persistent config management so past scans and tuning parameters could be reused instead of rerunning the full pipeline or hand-editing Python constants; used the dashboard to isolate failure modes around detection confidence, stereo matching, and calibration.

Autonomous Grocery Packing Robot

UCLA MAE 162D/E Capstone

- Owned the complete autonomy and embedded integration stack for a SCARA-like RRPR robot: stereo camera, overhead camera, Teensy 4.1 motion controller, stepper axes, and servo gripper, all connected through a Python-to-firmware command pipeline over serial.
- Chained YOLO segmentation, RAFT-Stereo disparity, camera-to-robot calibration, inverse kinematics, and 2.5D packing planning into one closed-loop system; achieved 95.1% pick-and-place success across 425 physical attempts.
- Built missed-pick recovery using post-grasp continuity checks and INA219 current sensing on the gripper to detect grasp/contact state directly from hardware, independent of the vision pipeline.

ADDITIONAL EXPERIENCE

UCLA Recreation / John Wooden Center

Summer 2025 – Jun. 2026

FITWELL Supervisor (promoted from Consultant)

Los Angeles, CA

- Coordinated daily operations, equipment maintenance tickets, and staff/patron communication in a high-traffic university facility; trusted with supervisor responsibilities during senior year.